Heckscher-Ohlin Model III

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Factor Price Equalization (FPE) Theorem

Factor Price Equalization (FPE) Theorem (developed by Paul Samuelson, 1948) argued that **free trade in goods alone** (without international factor mobility) will lead to complete equalization of **real factor prices** (wages and capital rents) `across countries`, even though factors themselves cannot move.

Core Idea: free trade in goods can substitute for international factor mobility. In simple terms: `Trade in goods acts as a substitute for trade in factors`. However, because many strong assumptions are required, FPE `rarely holds` fully in practice.

Core Assumptions (Same as H-O Model)

Assumption	Description
Two countries, Two goods, Two factors	Standard 2×2×2 model
`Perfect Competition`	In both goods and factor markets
Constant Returns to Scale (CRS)	Production functions are homogeneous of degree 1
Identical Homothetic Preferences	Same tastes across countries
Technology	Same technology in both countries
Factors mobile within country	Labor and Capital mobile between sectors
Factors immobile between countries	No international factor mobility
No transport costs / trade barriers	Free trade
Full employment	All factors are fully utilized

Main Mechanism

1. Before trade **(Autarky)**:
   - Labor-abundant country has **low wage (w)** and **high rental rate (r)**.
   - Capital-abundant country has **high wage (w)** and **low rental rate (r)**.
2. When **trade opens**:
   - Labor-abundant country exports labor-intensive good → increases demand for labor → **w rises**, **r falls**.
   - Capital-abundant country exports capital-intensive good → increases demand for capital → **r rises**, **w falls**.
3. `Goods prices` equalize across countries due to `free trade`.
4. Because `technology is identical` and `goods prices` are equal, cost minimization conditions force:

$$\frac{w}{r} = \frac{w^*}{r^*}$$

→ `Real factor prices (w and r)` become equal in both countries.

1. `Before trade`: Different relative endowments → different relative factor prices (w/r differs).
2. `Trade` opens → countries export goods using their `abundant factor` → relative goods prices `converge`.
3. Goods price equalization → identical `zero-profit` lines in both countries.
4. Since the unit input requirements (a_{ij}) depend on factor prices, the only solution consistent with both equations is **equal factor prices**

Key Insight: **FPE** is a **price transmission result**. Trade equalizes output prices; technology links output prices to input prices; thus, input prices equalize.

Important Implication

1. **Benchmark, Not Prediction**: FPE serves as a **theoretical baseline** to diagnose why convergence fails (e.g., tech gaps, trade frictions, institutional barriers).
2. **Regional Integration**: FPE holds more strongly within `integrated markets` (EU single market, US interstate trade, NAFTA/USMCA supply chains).
3. **Development Strategy**: Highlights that trade alone **cannot** equalize `living standards` without technology transfer, human capital investment, and institutional reform.
4. **Global Value Chains**: Task-based trade allows partial factor price convergence for specific skills/tasks, even if absolute wages diverge.
5. **Migration vs. Trade Policy**: If FPE held perfectly, migration restrictions would be unnecessary. Its failure explains why labor mobility remains politically and economically relevant.

Key Takeaways

1. FPE is a **logical consequence** of the H-O assumptions, not an empirical law.
2. It relies critically on **incomplete specialization** and the **diversification cone**.
3. The **magnification effect** (SS) and **output reallocation** (Rybczynski) are the adjustment mechanisms that enable FPE.
4. Empirical failure of FPE reveals the importance of technology, institutions, trade costs, and factor heterogeneity.
5. Modern trade theory uses FPE as a **counterfactual benchmark** to measure frictions, assess convergence, and design integration policies.

Link to Previous Theorems

Theorem	What Changes	What is Equalized / Affected	Direction
Stolper-Samuelson	Goods prices	Factor prices (magnified)	Opposite directions
Rybczynski	Factor endowments	Outputs (magnified)	Opposite directions
Factor Price Equalization	Goods prices (via trade)	**Factor prices across countries**	Complete equalization

- FPE is the **international counterpart** of Stolper-Samuelson: once goods prices are equalized, factor prices must also equalize.
- It explains why trade can substitute for factor mobility.

Why Factor-Price Equalization Often Fails in Reality

Here are the **main reasons** FPE does not hold in the real world:

Reason	Explanation	Effect on FPE
Different Technology	Countries have different production functions	FPE fails completely
Factor Intensity Reversal	Which good is labor-intensive changes with wage levels	FPE breaks down
Transportation Costs & Trade Barriers	Goods prices do not fully equalize	Partial or no equalization
More than Two Factors	Skilled labor, unskilled labor, land, etc.	Makes equalization very difficult
Specialization (Outside Cone)	Countries completely specialize	FPE does not occur
Non-homothetic Preferences	Different demand patterns	Affects goods prices
Imperfect Competition	Monopolies, unions, minimum wages	Prevents factor price adjustment
Political & Institutional Factors	Labor laws, unions, immigration restrictions	Blocks wage convergence
Capital Mobility	Capital moves easily, labor does not	Partial equalization only

Mathematical Derivation

1. Zero-Profit Conditions

$$P_X = a_{L_X} w + a_{K_X} r, \qquad P_Y = a_{L_Y} w + a_{K_Y} r$$

With `CRS` and `cost minimization`, $a_{i_j} = a_{i_j}(w/r)$. **At constant goods prices**, w/r is pinned down.

2. Invertibility Condition

Write in matrix form:

$$\begin{bmatrix} P_X \\ P_Y \end{bmatrix} = \begin{bmatrix} a_{L_X} & a_{K_X} \\ a_{L_Y} & a_{K_Y} \end{bmatrix} \begin{bmatrix} w \\ r \end{bmatrix}$$

The mapping from (P_X, P_Y) to (w,r) is `invertible` iff the `determinant` is non-zero:

$$\Delta = a_{L_X} a_{K_Y} - a_{L_Y} a_{K_X} \neq 0$$

This holds exactly when **no factor intensity reversal** occurs (i.e., X is consistently more/less labor-intensive than Y across all w/r).

3. Equalization Result

If $P_X^H = P_X^F$ and $P_Y^H = P_Y^F$, and `technologies are identical`, then:

$$\begin{bmatrix} w_H \\ r_H \end{bmatrix} = A^{-1} \begin{bmatrix} P_X \\ P_Y \end{bmatrix} = \begin{bmatrix} w_F \\ r_F \end{bmatrix}$$

Absolute FPE holds. Relative factor price equality $(w/r)_H = (w/r)_F$ follows immediately.

Under `free trade`, **identical** `production technologies`, and `incomplete specialization`, relative and absolute factor prices will equalize across countries, even if factors are internationally immobile.

In the 2×2×2 framework: If Home and Foreign trade freely in X and Y, and both produce both goods, then:

$$w_H = w_F, \quad r_H = r_F$$

- Trade in goods ≡≡ implicit trade in factor services.

1. Zero-Profit Conditions (Same in Both Countries)

Because technologies are identical and markets are competitive:

Home:

$$
\begin{aligned}
w_H a_{LX} + r_H a_{KX} = P_X \\
w_H a_{LY} + r_H a_{KY} = P_Y
\end{aligned}
$$

Foreign:

$$
\begin{aligned}
w_F a_{LX} + r_F a_{KX} = P_X \\
w_F a_{LY} + r_F a_{KY} = P_Y
\end{aligned}
$$

Since `free trade` equalizes goods prices (P_X and P_Y are the **same** in both countries), both countries face the **identical system** of two equations with two unknowns (w, r).

2. Matrix Form

$$\mathbf{A}' \begin{bmatrix} w \\ r \end{bmatrix} = \begin{bmatrix} P_X \\ P_Y \end{bmatrix}$$

Because the `input coefficient matrix` $\mathbf{A}$ is the same in both countries (`identical technology` + `same goods prices` → `same factor prices` from cost minimization), it follows that:

$$\begin{bmatrix} w_H \\ r_H \end{bmatrix} = \begin{bmatrix} w_F \\ r_F \end{bmatrix}$$

or `in percentage change` form (when goods prices change):

$$\begin{bmatrix} \hat{w} \\ \hat{r} \end{bmatrix} = (\mathbf{\Theta})^{-1} \begin{bmatrix} \hat{P}_X \\ \hat{P}_Y \end{bmatrix}$$

The right-hand side is identical for both countries.

Conditions Required for FPE

1. **Identical technologies** across countries.
2. **No factor intensity reversals** (factor intensity ranking remains the same at all factor prices).
3. **Both countries produce both goods** (they remain in the **cone of diversification** — no complete specialization).
4. **Free trade** with no transport costs or trade barriers.
5. **Perfect competition** and constant returns to scale.
6. Same number of factors and goods (or at least as many goods as factors).

Conditions for FPE to Hold

FPE holds only if:

- `Identical technology`; No factor intensity reversal
- Both countries continue producing **both goods** (diversification)
- Free trade with no transport costs; Same homothetic preferences

In reality, **most of these assumptions are violated**, which is why we do **not** observe full equalization of wages and capital returns between rich and poor countries.

Stolper-Samuelson and Rybczynski theorems examine `within-country` adjustments, FPE addresses `cross-country` convergence: it shows that, under strict conditions, free trade in **goods** can perfectly substitute for **international factor mobility**, equalizing wages and returns to capital across nations.

Historical Context & Core Statement

- **Origin**: Paul Samuelson (1948), *"International Trade and the Equalization of Factor Prices"* (Economic Journal).
- **Purpose**: To demonstrate the full general equilibrium implications of the H-O model when goods prices converge through trade.

Leontief Paradox

The **Leontief Paradox （1953)** is one of the most famous `empirical` challenges to the **Heckscher-Ohlin (H-O) model**. Wassily Leontief tested the H-O theorem using 1947 U.S. trade data. According to H-O, the United States — the most **capital-abundant** country in the world — should export **capital-intensive** goods and import **labor-intensive** goods. **Empirical Result**:

- U.S. **exports** were more **labor-intensive** than U.S. **imports**.
- Specifically, U.S. imports were about **30% more capital-intensive** than U.S. exports.

Core Idea: It showed that reality is more complex than “countries export goods that intensively use their abundant factors.”

- It opened the door to richer theories that include `human capital`, `technology gaps`, and `increasing returns to scale`. The empirical result was a **direct contradiction** of the H-O prediction → hence called a “paradox.”

Key Empirical Studies

Year	Researcher	Country	Main Finding	Supports H-O?
1953	Wassily Leontief	USA (1947)	Exports more labor-intensive than imports	**No**
1971	Robert Baldwin	USA (1962)	Imports 27% more capital-intensive	**No**
1980	Edward Leamer	USA	Paradox disappears with better methodology	Mixed
1987	Bowen, Leamer & Sveikauskas	27 countries	Mixed results; H-O rejected in many cases	Weak
2005	Kwok & Yu	USA	Paradox reduced or disappears in updated data	Improved
Recent	Various	USA	Paradox still appears, especially in services	Mixed

Main Explanations for the Leontief Paradox

Several attempts have been made to resolve the paradox:

1. `Higher U.S. Labor Productivity` (Leontief's own explanation)
    - American workers were more productive (better technology, education, management).
    - When adjusted for **productivity-equivalent labor**, the paradox disappears.
2. `Factor Intensity Reversal`
    - The assumption that a good is always labor- or capital-intensive may not hold across countries.
3. `Human Capital / Skilled Labor`
    - U.S. exports are intensive in **skilled labor** (human capital), not raw labor. Standard H-O only considers unskilled labor and physical capital.

4. `Natural Resources`
    - Many U.S. imports are resource-intensive (oil, minerals), which are capital-intensive to extract.
5. `Methodological Issues`
    - Leontief compared gross exports and imports instead of net trade.
    - Did not account for tariffs and trade barriers.
6. `Technology Differences`
    - H-O assumes identical technology across countries, which is unrealistic.

Current Status (Modern View)

- The strict version of the H-O model is **not strongly supported** by empirical evidence.
- The Leontief Paradox **still appears** in many studies, especially for the United States.
- However, when researchers include **human capital**, **technology differences**, or use **value-added trade** data, the paradox is **significantly reduced** or disappears in many cases.
- Modern trade theory has moved beyond simple H-O toward models that incorporate **technology**, **economies of scale**, **product differentiation**, and **institutions**.